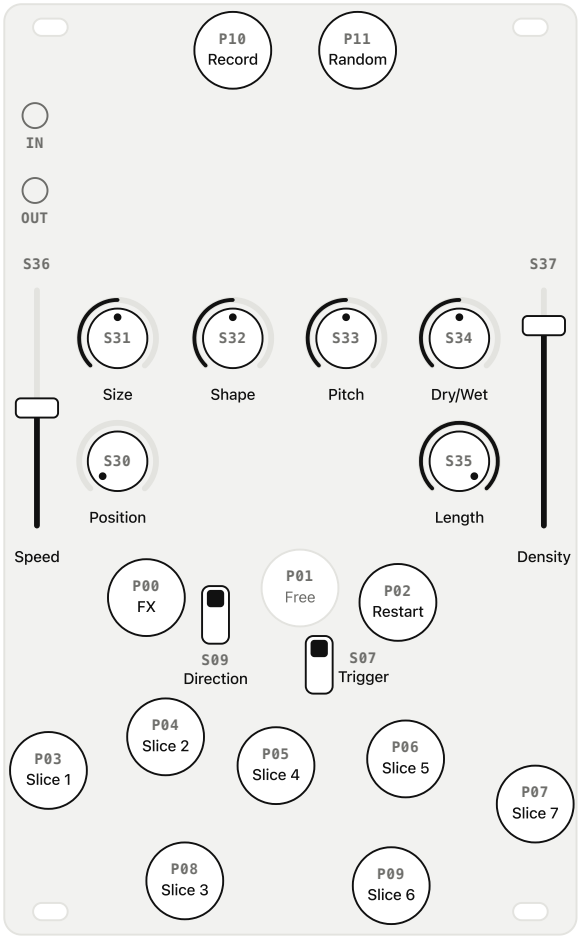
Simple Touch

Clessidra

It's the instrument where you pour grains of sound from single drops into a dense cloud.

Firmware and play in the browser: francescodimaggio.nl/clessidra

Made during the Synthux Design Residency 2026



Three layers on the knobs: their own, P11 held for randomness, P00 held for FX.

A knob takes over a value only once it reaches it, so nothing jumps.

Randomness: off at the centre, up an even spread, down mostly near with a long tail.

LAYERS

CONTROL	OWN	P11 RANDOM	P00 FX
S36	Speed	–	–
S30	Position	Position Random	–
S35	Length	–	–
S37	Density	–	–
S31	Size	Size Random	Filter
S32	Shape	Shape Random	Reverb
S33	Pitch	Pitch Random	–
S34	Dry/Wet	–	–

FADERS AND KNOBS

S36	How fast the playhead moves, 0 to 4×, 1× at the centre, a tape stop at the bottom.
S30	Where the selection sits, from the recording's start to its end.
P11 + S30	Randomness of position per grain: off at the centre, up an even spread, down a long tail.
S35	How much of the recording the selection holds, all of it at the top.
S37	Grains per second, 0.3 to 131. Periodic below the centre, off at it, random above.
S31	Grain length, 10 ms to 4 s, at most half the recording. 200 ms at the centre.
P11 + S31	Randomness of size per grain: off at the centre, up an even spread, down a long tail.
P00 + S31	Low-pass to the left, 20 kHz to 200 Hz, off at the centre, high-pass to the right, to 2 kHz.
S32	The grain window: reverse percussive, Hann at the centre, percussive.
P11 + S32	Randomness of shape per grain: off at the centre, up an even spread, down a long tail.
P00 + S32	Reverb on the grains: more and longer, 0.5 to 16 s.
S33	±2 octaves, stops at 0, 0.1, 1, 3, 4, 7, 12 and 24 semitones.
P11 + S33	Randomness of pitch per grain: off at the centre, up an even spread, down a long tail.
S34	The input and the grains, equal power. 0 % dry, 100 % wet.

PADS AND SWITCHES

P10	Tap: the last 8 s. Hold: until you let go, up to 30 s. Always listening.
P11	Held: S30 to S33 set the randomness of position, size, shape and pitch. Knobs pick up.
P00	Held: S31 sets the filter, S32 the reverb. Each knob picks up its value.
P02	The playhead from the start of the selection.
P03–P09	Plays its slice with its own grains, as Trigger sets. Press harder for more.
S09	Up forward, centre freeze, down reverse, for the playhead and the triggers.
S07	A slice pad plays once to its end (up), while held (centre), or looped while held (down).

LED

Glow	Grains playing
Flashing	Recording
Pulse	A control at or turning within its centre: pitch unison, speed 1×, size at its centre, density off
Steady	P11 or P00 held: the value of the knob last turned